Table 2: Comparison of recommendation performance across different models. MRR indicates the mean reciprocal rank and Hit@K measures proportion of relevant items in top-K results. QU and RG indicate query understanding and response generation capabilities, respectively.

| Models | Retrieval Methods | QU | RG | MRR(↑) | Hit@1(↑) | Hit@100(↑) |
|---|---|---|---|---|---|---|
| *Zero-shot Recommendation Models* | | | | | | |
| BM25 [40] | Sparse | ✓ | ✗ | 0.030 | 0.002 | 0.265 |
| CLAP-Music [49] | Similarity-based | ✓ | ✗ | 0.011 | 0.003 | 0.153 |
| NV-Embed-V2 [29] | Similarity-based | ✓ | ✗ | 0.029 | 0.005 | 0.255 |
| *Finetuned Recommendation Models* | | | | | | |
| CLAP-Music [49] | Similarity-based | ✓ | ✗ | 0.017 | 0.005 | 0.189 |
| NV-Embed-V2 [29] | Similarity-based | ✓ | ✗ | 0.032 | 0.005 | 0.281 |
| SASRec [26] | Generative | ✗ | ✗ | 0.002 | 0.000 | 0.008 |
| TIGER [39] | Generative | ✗ | ✗ | 0.008 | 0.001 | 0.092 |
| TALKPLAY (Ours) | Generative | ✓ | ✓ | **0.049** | **0.026** | **0.288** |

# 5 Results

## 5.1 Conversational Music Recommendation Results

Our results in Table 2 demonstrate that TALKPLAY significantly outperforms both zero-shot and finetuned baseline models across all key metrics. TALKPLAY achieves an MRR of 0.049, which is higher than the best baseline (finetuned NV-Embed-V2 at 0.032). More notably, TALKPLAY shows a dramatic improvement in Hit@1 performance (0.026), achieving more than 5 times the accuracy of the next best model. This demonstrates a strong correlation between the cross-entropy loss function used in next token prediction and the Hit@1 metric. The superior Hit@1 performance of TALKPLAY suggests that our generative approach can directly produce highly relevant recommendations without requiring extensive candidate consideration in the realistic music listening experience.

Among zero-shot recommendation models (upper side of Table 2), traditional sparse retrieval methods such as BM25 and embedding-based approaches like NV-Embed-V2 exhibit comparable performance, with MRR scores of 0.030 and 0.029 respectively. This indicates that, in the absence of task-specific training, dense embedding method [29] offer limited benefit over simpler retrieval techniques [40]. One reason for this close performance is that a large portion of music-related queries involve metadata—such as track titles, artist names, and album information—where sparse retrieval methods like BM25 are particularly effective due to their strength in exact token matching. In contrast, CLAP [49] performs significantly worse in multi-turn recommendation scenarios (MRR of 0.011). Given that CLAP was trained primarily on semantic annotations like genre and instruments without conversational context, this suggests its limited dialogue understanding capability leads to lower performance in conversation-based recommendation scenarios.

In fine-tune recommendation models (lower side of Table 2), fine-tuning improves performance for all baseline models, but the gains are modest compared to TALKPLAY's results. Notably, the sequencial recommendation models, such as SASRec [26] and TIGER [39], perform significantly worse than retrieval-based approaches. It indicates that the challenge of generating accurate recommendations without query understanding capabilities in conversational recommendation scenario.

Figure 2 shows how retrieval performance varies across conversation turns. The MRR scores reveal an interesting pattern: while TALKPLAY starts with comparatively lower scores in the first 1-2 turns, it significantly outperforms all baselines from Turn 4 onward and maintains relatively stable performance even as the conversation extends to later turns to Turn 13. In contrast, baseline methods exhibit notable degradation as the conversation continues. This pattern shows a fundamental trade-off in conversational recommendation

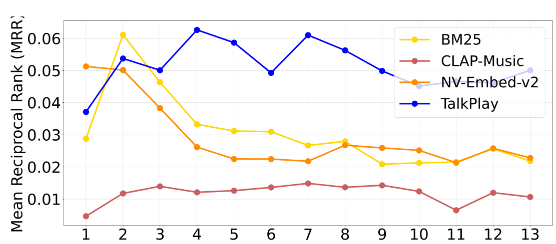


Figure 2: Performance comparison across conversation turns. The x-axis shows the turn number in the dialogue, and the y-axis shows Mean Reciprocal Rank (MRR).

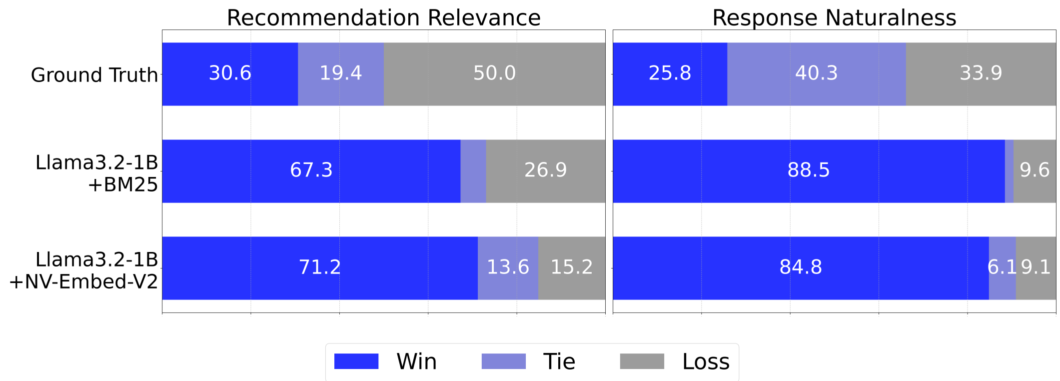


Figure 3: A-vs-B human evaluation results, comparing TALKPLAY against existing conversational music recommendation models on recommendation relevance (left) and response naturalness (right).

systems: TALKPLAY requires a few dialogue exchanges to establish user preferences and context, but once established, it excels at maintaining this understanding throughout extended interactions.

This result also emphasizes the fundamental challenge of multi-turn recommendation. Similarity-based retrieval methods must rely on simple aggregation strategies (e.g., averaging) to combine the previous chat history, which becomes less effective as context grows more complex. In contrast, TALKPLAY's LLM-based approach leverages the model's internal attention mechanism during token prediction to naturally capture and prioritize relevant contextual signals throughout the conversation. This enables more nuanced understanding of evolving user preferences, particularly beneficial in real-world music discovery scenarios where conversations typically extend beyond a few initial exchanges.

### 5.2 Qualitative Results

Figure 3 presents the results of our A-vs-B human evaluation, which compares TALKPLAY against both ground truth data and a set of baseline models. We conducted a human evaluation study to assess the quality of our model's recommendations and responses. Participants were presented with multi-turn conversation scenarios and asked to compare recommendations and generated responses between two models. The survey interface is showed in the appendix C.

In the comparison with ground truth data,TALKPLAY demonstrates competitive performance: it achieves a combined *Win+Tie* rate of 50% for the *Recommendation Relevance* metric and 66.1% for *Response Naturalness*. These scores indicate that the outputs generated by TALKPLAY are on par with ground truth recommendations and responses in terms of both relevance and fluency.

When compared against baseline models, TALKPLAY exhibits a clear performance advantage. Specifically, in the domain of recommendation relevance, our model outperforms both baselines: Llama3.2-1B + NV-Embeds-V2 with a Win Rate of 71.2% and Llama3.2-1B + BM25 with a Win Rate of 67.3%. These results validate the superiority of our LLM-based approach over traditional retrieval-based baselines in multi-turn recommendation scenario.

The improvements are even more substantial in the response generation task, underscoring the proposed model's capability to generate more natural and contextually appropriate replies within music-related conversations. Human evaluators consistently preferred our model's responses, with Win Rates of 84.8% and 88.5% over those generated by the general-purpose Llama-3.2-1B Instruct model.

### 5.3 Multimodal Feature Importance

Tables 3 and 4 present comprehensive ablation studies that systematically analyze the relative importance of different modalities and their weighting strategies in our model. Our analysis of weighting strategies in Table 3 reveals a clear pattern: approaches that emphasize coarse-grained features (playlist and semantic tokens) over fine-grained features (lyrics and audio tokens) consistently demonstrate superior performance. Specifically, both linear and quadratic coarse-to-fine weighting schemes achieve an MRR of $0.049$, representing a substantial 29% improvement over the uniform

Table 3: Ablation study results showing the impact of different weighting strategies for modality tokens in TALKPLAY. The modality weights are applied in the order: (playlist, metadata, attribute, lyrics, audio).

| Weight Methods | Modality Weight | MRR ↑ | Hit@1 ↑ | Hit@10 ↑ | Hit@100 ↑ |
|---|---|---|---|---|---|
| Uniform | (1, 1, 1, 1, 1) | 0.038 | 0.023 | 0.066 | 0.180 |
| Linear (fine-to-coarse) | (1, 2, 3, 4, 5) | 0.025 | 0.018 | 0.037 | 0.085 |
| Quadratic (fine-to-coarse) | (1, 4, 9, 16, 25) | 0.029 | 0.020 | 0.043 | 0.124 |
| Linear (coarse-to-fine) | (5, 4, 3, 2, 1) | 0.049 | 0.027 | 0.090 | 0.283 |
| Quadratic (coarse-to-fine) | (25, 16, 9, 4, 1) | 0.049 | 0.026 | 0.096 | 0.288 |

weighting baseline (0.038). This performance gap strongly suggests that higher-level contextual information encoded in playlists and semantic descriptors serves as a more reliable signal for conversational recommendation compared to granular feature-level information.

Table 4: Ablation study results showing the impact of removing different modalities from TALKPLAY.

| Models | MRR | Delta |
|---|---|---|
| TALKPLAY $_{\text{uniform}}$ | 0.0376 | - |
| remove Playlist Token | 0.0281 | −0.0096 |
| remove Metadata Token | 0.0337 | −0.0039 |
| remove Attribute Token | 0.0372 | −0.0005 |
| remove Lyrics Token | 0.0374 | −0.0003 |
| remove Audio Token | 0.0407 | +0.0036 |

The leave-one-out experiments presented in Table 4 provide further empirical support for this finding. The ablation of playlist tokens results in the most significant performance degradation (−0.0096 MRR), followed by metadata tokens (−0.0039 MRR). This hierarchical impact aligns with our theoretical understanding that playlist co-occurrence and metadata capture essential contextual signals about music similarity and user preferences. A particularly noteworthy observation is that removing audio tokens leads to a slight performance improvement (+0.0036 MRR). This counter-intuitive result suggests that the low-level audio features used in TALKPLAY may actually introduce noise in conversational recommendation scenarios, where user preferences are predominantly articulated through natural language rather than acoustic similarity.

These empirical findings carry important implications for the design of future conversational music recommendation systems. First, they demonstrate that diverse multimodal representations can enhance recommendation performance. While different features may contribute differently depending on the conversation context, our results suggest that coarse-grained features such as playlist and metadata tokens play a crucial role in recommendation quality. As a future direction, we suggest analyzing the relative importance of different modalities based on specific conversation characteristics to further optimize recommendation systems.

## 6 Conclusion

We presented TALKPLAY, demonstrating that music recommendation can be effectively reformulated as a token generation task within a large language model (LLM) framework. This novel perspective simplifies the architecture of recommendation systems by unifying dialogue understanding and recommendation generation into a single model. Our key contributions, the generation of synthetic conversations for training, a multimodal tokenization strategy, and a unified conversational recommendation model—collectively enable stable end-to-end training while delivering competitive performance. Notably, this approach reduces system complexity by eliminating the need for separate components such as dialogue managers or retrieval-based ranking modules. Empirical results indicate that TALKPLAY maintains high recommendation quality across different modalities, including text and audio, highlighting its generalizability and robustness.

Looking ahead, there are several promising directions for future research. Scaling the model to larger and more diverse datasets could further enhance its recommendation accuracy and cultural relevance. Incorporating additional modalities—such as visual content or user behavior logs—may improve contextual understanding and personalization. Moreover, leveraging LLM-specific capabilities, such as chain-of-thought reasoning, zero-shot generalization, and in-context learning, offers exciting opportunities to push the boundaries of conversational recommendation systems.